INSTRUCTOR KEY: LAUREN MITCHELL TEAM CASE STUDY

about 2 hours

What this is

A 2-hour team case study. The whole class (designed for 7 students) works as ONE team and cares for Lauren Mitchell (46, adrenal crisis in known Addison disease after a stomach virus) across 24 hours of patient time, **Mon 0700 to Tue 0700**. The class advances **4 hours together** (0700, 1100, 1500, 1900, 2300, 0300, then a final check at 0700). The packet is 12 pages: cover, patient card with a hormone map and two EHR screenshots, the rubric with a **response tracker**, one page per checkpoint, two final pages (evaluation, then teaching and discharge) and an individual reflection.

Focus: pathophysiology (cortisol and aldosterone missing: low sodium, high potassium, low volume, low glucose; high ACTH: dark skin), assessment, the standard treatment plan (stress-dose hydrocortisone on time, fluids, dextrose, fludrocortisone held then restarted), and **evaluation** of the response (the tracker every stop and the evaluation table at the end).

Roles and pairs (7 students, one team)

- **Charge nurse** (rotates; each student has exactly one stop): runs the section, keeps time, **gets the findings from each pair when the pair is finished**, decides the top 3 priorities (writes their name in that box), then develops and gives the SBAR. The questions are NOT printed for students: they learn to ask them. The "questions a strong charge nurse asks" in each section below are for you to prompt with only if a team is stuck.
- **Meds & Treatments pair:** MAR, orders, notes: what is running, due, held or should change, and why.
- **Vitals & Assessment pair:** read the bedside monitor and write the vitals row, orthostatic vitals, the bedside exam (mental status, skin, mouth, belly), intake and output. **Fills the response tracker column at every stop.**
- The **Vitals & Assessment** and **Meds & Treatments** pairs work together on "Actions to suggest" at every stop. Every student has their own packet, writes their name in the header and writes their pair's names in their pair box.
- **Results & Labs pair:** new labs, ECGs, trends and what they mean (sodium rate, potassium, glucose, cortisol/ACTH).
- **Pairs switch jobs at every checkpoint.** With 7 students: 1 charge nurse + 3 pairs. You (or the charge nurse) assign the jobs at the start of each stop; students write their names in their pair box. Each student is charge nurse once (7 stops).
- At each stop the old charge nurse takes the new charge nurse's place in that pair.

Before class

- Print **one packet per student** (7), single-sided. Each student writes their name in the header; the last page is their individual reflection and peer check.
- Load the EHR: Pre-Built Patients > Case Study Patients > Lauren Mitchell (or Import with casestudy/cases/addison/patient.json). Check that the clock reads **Mon 07:00** (4W-22, SIM-40422).
- Open the bedside monitor (Monitor button) in the same browser; it loads Lauren's built-in scenario by MRN and follows the 4h button. The monitor BP is **lying**: you say the standing values aloud.
- **There are no vital signs or bedside glucoses in the chart after 0700.** You give the glucose and the standing values aloud (in each checkpoint section below). Ignore the instructor-only "Suggested vitals" bar on the Flowsheets page (it repeats the 0700 set).
- Optional moulage: bronzer on the face, neck and backs of the hands, darker palm creases; sweat spray at 1100.

Rules that protect the pacing

- Students finish a page at the **end** of its time block. Nothing on a later page is printed on an earlier one (checked by check_facts.py). Do not let anyone turn the page until you advance the clock.
- Every number in this key matches the chart (patient.json), the instructor guide and the monitor. If the team and the key disagree, check the chart.
- Answers can differ from the model. Look for chart numbers and a reason, not exact wording.

Run of show (about 2 hours)

Each checkpoint follows the same 13-14 minute rhythm. The final 0700 gets 17 minutes because it has two pages.

Clock	Minutes	What the instructor does
Start	0:00-0:05	Welcome. Everyone sits together as one team; pick the 0700 charge nurse and assign the three pair jobs (they switch every stop). Read the quick pathophysiology on the cover together. Walk through the patient card and hormone map (page 2), the rubric and the response tracker (page 3). Student A is charge nurse at 0700.
0700	0:05-0:19	Do NOT press 4h (the 0700 set is in the chart). Read the situation. The charge nurse gets each pair's findings; the team also fills the hormone map (page 2) and the 0700 tracker column; SBAR call to you (warm-up, not scored).
1100	0:19-0:33	Press **4h** (Mon 1100). New charge nurse; pairs switch jobs. Give the glucose **58** when they check. Watch for juice vs D50 and for checking the 1000 hydrocortisone.
1500	0:33-0:47	Press **4h** (Mon 1500). New charge nurse; pairs switch jobs. Give the standing values when they do orthostatics. Watch for the sodium calculation (+6 in 9 h) and the call.
1900	0:47-1:01	Press **4h** (Mon 1900). New charge nurse; pairs switch jobs. The fluid change happened at 1530. Watch for evaluation with numbers ("is it working?").
2300	1:01-1:14	Press **4h** (Mon 2300). New charge nurse; pairs switch jobs. Watch for sedative requests and insulin for 152.
0300	1:14-1:27	Press **4h** (Tue 0300). New charge nurse; pairs switch jobs. Describe the infiltrated right forearm IV when they look. Watch for the plan for the 0400 IV dose.
Final 0700	1:27-1:44	Press **4h** (Tue 0700). New charge nurse; pairs switch jobs. Page 1: pairs, tracker, evaluation table and verdict. Page 2: handoff, steroid plan, sick-day rules, the injection (return demonstration with a training vial if you have one), ID, teach-back, readiness.
Reflection	1:44-1:50	Each student completes their own reflection and peer check (page 12). Collect the 7 packets.
Debrief	1:50-2:00	Use the debrief questions at the end of this key. Put the tracker on the board.

The same 14 minutes, one checkpoint

Minute	What happens
+0	Press 4h (except at the first 0700). Check the monitor moved to the new row. Read the situation aloud. Remind them the charge nurse changed.
+1 to +3	Everyone fills the situation blanks from the chart. Pairs decide what to look for (no questions are printed; prompt with the questions below only if a pair is stuck). Give glucose and standing values aloud when they ask or check.
+3 to +10	Pairs work the chart, the monitor and "Lauren", write in their box with their names, and report to the charge nurse when finished. Vitals & Assessment + Meds & Treatments agree on actions to suggest; Vitals & Assessment fills the tracker column. Ask a charge nurse "what is your team deciding?"
+10 to +13	The charge nurse decides the top 3 priorities, then gives you the SBAR. Answer as the provider with the suggested reply. Keep it short.
+13 to +14	Check that the tracker column and the prediction are written. Then advance.

Monitor and bedside values at a glance

Stop	T	HR	BP lying (MAP)	Standing (say aloud)	RR	SpO2	Pain	Rhythm	Glucose (say aloud)
Mon 0700	99.4	104	98/60 (73)	not safe to stand (dizzy sitting up)	18	98% RA	4/10	Sinus tachycardia, tall peaked T waves	88 (0645)
Mon 1100	99.0	112	94/58 (70)	not done (lying SBP under 100)	20	98% RA	3/10	Sinus tachycardia (T waves a little less tall)	58 (142 at 1120 after D50)
Mon 1500	98.8	92	108/66 (80)	94/60, HR 114, dizzy (POSITIVE)	16	99% RA	2/10	Normal sinus rhythm, normal T waves	118
Mon 1900	98.6	86	112/70 (84)	104/66, HR 96, no dizziness (negative)	16	99% RA	1/10	Normal sinus rhythm	142
Mon 2300	98.4	80	116/72 (87)	112/70, HR 88, no symptoms (negative)	16	98% RA	0/10	Normal sinus rhythm	152
Tue 0300	98.2	76	118/74 (89)	not repeated (just back to bed after the near-fall)	14	98% RA	0/10	Normal sinus rhythm (T waves a little smaller)	128

Instructor copy. Every number matches the chart (patient.json), the instructor guide and the monitor.

Tue 0700	98.2	74	120/76 (91)	116/74, HR 80 (negative)	14	99% RA	0/10	Normal sinus rhythm	112

0700 Monday: Checkpoint 1 of 6

Monitor (the NOW row): HR 104, BP 98/60 lying (MAP 73), RR 18, SpO2 98% room air, temp 99.4, pain 4/10 crampy belly; rhythm: Sinus tachycardia, tall peaked T waves. **Standing (aloud):** not safe to stand (dizzy sitting up). **Bedside glucose (aloud):** 88 (0645)

Urine and I&O for the Vitals pair: ED 0335-0700: in 2,700 IV (fluids 2,650 + D50 50) / out 500 (urine 150 at 0450, emesis 150, stool 200). First void on the unit is at 0800 (120 mL).

Portray: Drowsy, wakes to voice, oriented to person and place. "I'm so tired... my stomach hurts... I just want some pickles. Did I miss my pills?" If they sit her up: "the room is spinning". Daniel: "Nobody ever told us what to do. We don't have that shot."

Situation (blanks with answers in brackets): It is **Monday 0700**, 4 West, room 4W-22. Lauren Mitchell, age **[46]**, just came up from the ED. Her history: **[Addison disease (primary adrenal insufficiency)]**. Since Saturday she has had **[vomiting and watery diarrhea]** and could not keep down her **[hydrocortisone and fludrocortisone]** pills. EMS found her confused on the bathroom floor at 0318: BP **[74/40]**, glucose **[58]**. EMS and the ED gave **[2.5]** L of saline, dextrose, and hydrocortisone **[100 mg IV]** at **[0420]**. Running now: **[D5NS at 150]** mL/h. Her husband **[Daniel]** is at the bedside.

Questions a strong charge nurse asks (not printed for students; prompt only if stuck): Meds & Treatments: "Which steroid is due next, and when? Why are the home hydrocortisone and fludrocortisone held? What is running in the IV?" **Vitals & Assessment:** "What do the monitor and Lauren show? Skin color, mouth, mental status, belly. Is she safe to stand?" **Results & Labs:** "ED sodium, potassium, glucose, creatinine: what does each mean? What do the ECG T waves show? What is still pending?"

New results (chart values)

Test	Normal	Mon 0406
Sodium	136-145	126
Potassium	3.5-5.1	6.0
Glucose (serum)	70-110	49
BUN	7-20	38
Creatinine	0.5-1.1	1.6
CO2 (bicarbonate)	22-29	18
Calcium	8.6-10.2	10.6
Lactate	0.5-2.0	3.1
Eosinophils	0-5	7.0

Test	Normal	Mon 0450
Urine sodium	Under 20 expected when a person is dehydrated	48

Box	Model answer and rationale
MEDS & TREATMENTS pair	Hydrocortisone 50 mg IV every 6 hours: **first unit dose due 1000** (about 6 h after the ED 100 mg at 0420); time-critical, give within 30 minutes, never hold for a "good" BP. Home oral hydrocortisone held (she is on IV; tablets would not stay down). **Fludrocortisone held:** 200 mg/day of hydrocortisone already acts like aldosterone (over 50 mg/day). Levothyroxine held today (resumes Tue 0600). D5NS 150 mL/h in the right forearm 20 g; 18 g left forearm saline lock. PRN: ondansetron (last 0430), hypoglycemia protocol (15 g by mouth, or D50 25 g IV if she cannot swallow safely).
VITALS & ASSESSMENT pair	HR 104 sinus tach with tall T waves, BP 98/60 (MAP 73), RR 18, SpO2 98% RA, T 99.4, pain 4/10 crampy belly, glucose 88 (0645). Drowsy, oriented to person and place only, repeats questions (GCS 14). **Tan/bronze skin**, dark palm creases and knuckles, dark patches inside the cheeks. Dry mouth, skin tents, hands cool, cap refill 3 s. Bruised knees. Not safe to stand (dizzy sitting up). Urine: 150 mL dark at 0450 (ED); first void on the unit 0800.
RESULTS & LABS pair: new results and significance	Sodium 126 (low: aldosterone missing, salt lost in urine and stool; cortisol missing lets ADH hold water). Potassium 6.0 (high: no aldosterone to excrete it) with **peaked T waves V2-V4** on the 0425 ECG (PR 148, QRS 88: no severe signs). Glucose 49 (no cortisol). BUN 38 / creatinine 1.6 (baseline 0.8): dehydration, prerenal. CO2 18, lactate 3.1 (shock), Ca 10.6 (volume contraction), eosinophils 7.0%. **Urine sodium 48** while dehydrated = salt wasting. Pending: cortisol and ACTH drawn at 0406 before the steroid.
Response tracker column (Vitals & Assessment pair)	98/60 (73); standing not safe; 104, tall T; 88; Na 126; K 6.0; drowsy, x2; 150 mL in the ED, dark; nauseated; (start)
Actions to suggest (Vitals & Assessment + Meds & Treatments)	Hydrocortisone ready for 1000; glucose at 0900; BMP 0900; strict I&O with the hat; bed rest and bed alarm; stool sample ready; ask Daniel what they knew about sick days.

TEAM (charge): top 3 priorities	1) Perfusion and rhythm: BP/MAP, HR, telemetry (tall T waves, K 6.0), mental status, urine; IV sites working. 2) Hydrocortisone 50 mg IV at 1000 ON TIME (and glucose checks every 2 hours with the protocol ready). 3) Safety: bed rest, bed alarm, commode with assist (dizzy, confused); contact (enteric) precautions with soap and water.
TEAM (charge): SBAR (warm-up, not scored)	**S:** RN for Lauren Mitchell, 4W-22, just arrived from the ED in adrenal crisis. BP 98/60, HR 104 with tall T waves, drowsy and oriented x2. **B:** 46, Addison disease, vomited her hydrocortisone since Saturday. Na 126, K 6.0, glucose 49 in the ED; hydrocortisone 100 mg IV 0420, D50 and 2.5 L saline, then D5NS. **A:** Improving after the stress dose but still in crisis; at risk for low glucose, low BP and arrhythmia. **R:** Confirm hydrocortisone 50 mg IV at 1000 and the plan for glucose and BMP checks; call parameters please.
Prediction (no wrong answers)	Any reasoned guess earns credit. What happens: the 0900 labs improve, but at 1100 she has a second low glucose (58) with sweating.
Charge nurse notes	Look for: each pair was asked its question, the charge nurse decided, a time check. Credit any honest notes.

Hormone map (page 2): model answers for "What Lauren shows"

Hormone	What Lauren shows
CORTISOL (missing)	Low BP that responds poorly to fluid alone (EMS 74/40; 98/60 now), HR 104; glucose 49 (ED) and 58 at EMS; nausea, vomiting, belly pain, weakness, confusion (oriented x2); temperature 99.4; eosinophils 7.0%.
ALDOSTERONE (missing)	Sodium 126 with **urine sodium 48** while dehydrated (salt wasting); potassium 6.0 with peaked T waves; CO2 18; dehydration (dry mouth, skin tents) and prerenal kidney injury (BUN 38, creatinine 1.6); salt craving ("pickles"); dizzy sitting up.
ACTH (very high)	Tan/bronze skin in December, dark palm creases and knuckles, dark patches inside the cheeks. (ACTH result comes at 1440: 1,240.)

Instructor as the provider (suggested reply): Thanks. The 1000 hydrocortisone is time-critical: give it within 30 minutes of 1000. Keep D5NS at 150, glucose every 2 hours, BMP every 4 hours (sodium no more than 8 up in 24 hours), bed rest and telemetry. Call me for SBP under 90, glucose under 70 or any rhythm change. I am asking endocrinology and pharmacy to see her, and please send the first liquid stool for C. diff and a GI panel.

1100 Monday: Checkpoint 2 of 6

Monitor (the NOW row): HR 112, BP 94/58 lying (MAP 70), RR 20, SpO2 98% room air, temp 99.0, pain 3/10 belly; rhythm: Sinus tachycardia (T waves a little less tall). **Standing (aloud):** not done (lying SBP under 100). **Bedside glucose (aloud):** 58 (142 at 1120 after D50)

Urine and I&O for the Vitals pair: 0700-1100: in 660 (IV 600, PO 60) / out 630: voids 0800 120, 0930 140, 1050 120 (380 mL, about 95 mL/h) + liquid stool 250 at 0945.

Portray: Shaky, sweaty, anxious, foggy: "Something's wrong... I feel like I'm going to pass out." Give glucose **58**; if rechecked 15 minutes after D50: **142**. You may lower HR to 100 on the monitor after treatment.

Situation (blanks with answers in brackets): It is **1100**. The 0900 bedside glucose was **[72]**; the 0900 serum sodium is **[128]** and potassium **[5.5]**. Her hydrocortisone **[50 mg IV]** was due at **[1000]**. She had a large liquid stool at 0945 and the sample went to the lab. You walk in for the 1100 glucose check. Lauren is sweating and says, "Something's wrong." Her bedside glucose is **[58]** (ask the instructor). Her mental status compared with 0700: **[more foggy, oriented to person and place]**.

Questions a strong charge nurse asks (not printed for students; prompt only if stuck): Meds & Treatments: "Was the 1000 hydrocortisone given on time? What does the protocol say for a low glucose in a nauseated, foggy patient: which route, how much, when do we recheck?" **Vitals & Assessment:** "What is the glucose now and what symptoms do you see? Mental status compared with 0700? Urine and stool since 0700?" **Results & Labs:** "0900 labs against the ED labs: sodium, potassium, creatinine, glucose, lactate. Better or worse? Why can glucose fall even with D5NS running?"

New results (chart values)

Test	Normal	Mon 0406	Mon 0900
Sodium	136-145	126	128
Potassium	3.5-5.1	6.0	5.5
Creatinine	0.5-1.1	1.6	1.4
Glucose (serum)	70-110	49	74
CO2 (bicarbonate)	22-29	18	19
Lactate	0.5-2.0	3.1	1.4

Box	Model answer and rationale
MEDS & TREATMENTS pair	Hydrocortisone 50 mg IV due 1000: check the MAR that it was given on time (if late, this is the moment to connect it). Glucose 58 + nausea + "foggy" = **D50 25 g IV** (protocol), slowly through the 18 g left forearm IV after checking blood return (vesicant); **recheck in 15 minutes** (142 at 1120). Keep D5NS 150 mL/h running. Ondansetron 4 mg IV is available (last dose 0430) so she can start sips.
VITALS & ASSESSMENT pair	HR 112, BP 94/58 (MAP 70), RR 20, SpO2 98%, T 99.0, pain 3/10. **Glucose 58**: shaky, sweaty, anxious, cool moist skin, oriented to person and place (at 0900 she knew the date). Urine 0800 120, 0930 140, 1050 120 = 380 mL (about 95 mL/h), getting lighter. Liquid stool 250 mL at 0945. In 660 / out 630.
RESULTS & LABS pair: new results and significance	Sodium 128 (+2 in 5 h), potassium 5.5 (down from 6.0), creatinine 1.4, CO2 19, lactate 1.4 (down from 3.1): **improving**. Serum glucose 74 at 0900 hinted the fall. Why low with D5NS: no cortisol = the liver cannot make glucose, glycogen used up after 2.5 days of no food, nothing by mouth; hydrocortisone takes hours to fix this. D5NS at 150 mL/h gives only 7.5 g/h.
Response tracker column (Vitals & Assessment pair)	94/58 (70); not done; 112; 58 (142 after D50); Na 128; K 5.5; foggy, x2; about 95 mL/h; mild nausea; worse (glucose); labs better
What changed since the last stop	Worse: glucose 88 to 58, HR 104 to 112, BP 98/60 to 94/58, more confused. Better: labs (Na 126 to 128, K 6.0 to 5.5, creatinine 1.6 to 1.4, lactate 3.1 to 1.4).
Actions to suggest (Vitals & Assessment + Meds & Treatments)	D50 now, recheck at 1115-1120; glucose every hour; ondansetron; clear sips when nausea settles; hydrocortisone at 1600 on time.
TEAM (charge): top 3 priorities	1) Treat the low glucose now: D50 25 g IV, recheck in 15 minutes, keep the dextrose fluid running. 2) Make sure the hydrocortisone was on time (and stays on time at 1600); call the provider (SBAR). 3) Safety: fall precautions (shaky, confused), IV site check before D50, nausea control so she can start sips.
TEAM (charge): SBAR	**S:** Lauren Mitchell, 4W-22: glucose 58, shaky, sweaty and more confused; HR 112, BP 94/58. I am giving D50 25 g IV now per protocol. **B:** Addison disease in adrenal crisis; glucose 52 in the ED, 72 at 0900; D5NS 150 mL/h; nauseated, NPO. Hydrocortisone 50 mg IV given at 1000. **A:** Low glucose from cortisol deficiency and no food; the hydrocortisone has not had time to work. **R:** I will recheck in 15 minutes. Do you want more frequent glucose checks? May I give ondansetron and start sips? Please come and see her.
Prediction (no wrong answers)	She should be much better by 1500 (steroid working). What happens: alert x3, BP 108/66, but orthostatic and sodium rising too fast with a big urine output.

Instructor copy. Every number matches the chart (patient.json), the instructor guide and the monitor.

Charge nurse notes	Look for: each pair was asked its question, the charge nurse decided, a time check. Credit any honest notes.

Instructor as the provider (suggested reply): Good: give the D50 and recheck in 15 minutes. I will order glucose every hour twice, then every 2 hours until three readings over 70. Keep the D5NS running. Ondansetron is fine; sips when the nausea settles. Make sure the 1600 hydrocortisone is on time. I will come by.

1500 Monday: Checkpoint 3 of 6

Monitor (the NOW row): HR 92, BP 108/66 lying (MAP 80), RR 16, SpO2 99% room air, temp 98.8, pain 2/10; rhythm: Normal sinus rhythm, normal T waves. **Standing (aloud):** 94/60, HR 114, dizzy (POSITIVE). **Bedside glucose (aloud):** 118

Urine and I&O for the Vitals pair: 1100-1500: in 770 (IV fluid 600 + D50 50 charted as IV med, PO 120) / out 1,050: voids 1150 250, 1300 300, 1400 280, 1450 220 = about 260 mL/h of pale urine.

Portray: "I feel so much better. Can I go to the bathroom by myself? I have to go all the time." Orthostatics aloud: lying 108/66 HR 92; standing at 1 minute **94/60, HR 114**, "I'm dizzy, I need to sit down." Lay her back down.

Situation (blanks with answers in brackets): It is **1500**. Lauren has kept clear liquids down since **[1345]**, and endocrinology saw her. The ED cortisol (before the steroid) is **[1.8]** and the ACTH is **[1,240]**. The 1300 sodium is **[132]**: up **[6]** from 126 in **[9]** hours. The urine hat keeps filling with pale urine: about **[1,050]** mL since 1100. Her lying BP is over 100 and she is alert, so **orthostatic vital signs are due**. The next hydrocortisone is due at **[1600]**.

Questions a strong charge nurse asks (not printed for students; prompt only if stuck): Meds & Treatments: "Hydrocortisone is due at 1600 and her BP is better: give it or hold it? Why? Which IV fluid is running and at what rate? What change would you ask for?" **Vitals & Assessment:** "Orthostatic vitals, done safely: lying, then standing at 1 minute. Positive or negative? Urine per hour since 1100? Mental status?" **Results & Labs:** "Sodium: 126 at 0406 to ___ at 1300 = ___ in ___ hours. The limit is 8 in 24 hours: on track or too fast? Cortisol and ACTH: what do they mean together? 1330 ECG T waves?"

New results (chart values)

Test	Normal	Mon 0406	Mon 0900	Mon 1300
Sodium	136-145	126	128	132
Potassium	3.5-5.1	6.0	5.5	4.9
Creatinine	0.5-1.1	1.6	1.4	1.2
Glucose (serum)	70-110	49	74	96

Test	Normal	Mon 0406
Cortisol (random, before steroid)	Morning 5-23; in severe illness expected over 18	1.8
ACTH (before steroid)	7.2-63	1240

Box	Model answer and rationale
MEDS & TREATMENTS pair	**Give the 1600 hydrocortisone on time**: the steroid is WHY she looks better; holding it brings the crisis back. Fludrocortisone still held (200 mg/day covers aldosterone). D5NS 150 mL/h is running and adds sodium: ask for a change. What the provider orders at **1530: stop D5NS, start D5 0.45% NaCl 100 mL/h** + a sodium-limit order (134 or lower until Tue 0400). Hang it on time and relabel the line.
VITALS & ASSESSMENT pair	HR 92, BP 108/66 lying (MAP 80), RR 16, SpO2 99%, T 98.8, pain 2/10, glucose 118, normal sinus rhythm with normal T waves. **Orthostatic: standing 94/60, HR 114, dizzy = POSITIVE** (SBP -14, HR +22, with symptoms). Alert, oriented x3, nausea gone. Urine 1150 250, 1300 300, 1400 280, 1450 220 = 1,050 mL in 4 h, **about 260 mL/h pale** (water diuresis). In 770 / out 1,050.
RESULTS & LABS pair: new results and significance	**Sodium 126 to 132 = +6 in 9 hours**: at this pace she will pass +8 in 24 h tonight (too fast: risk of osmotic demyelination). Potassium 4.9, creatinine 1.2, glucose 96. **Cortisol 1.8** (critical low; in severe illness it should be over 18) with **ACTH 1,240** (normal 7.2-63) = **primary** adrenal insufficiency: the gland cannot answer, so the pituitary shouts; high ACTH explains the tan skin and the need for fludrocortisone. ECG 1330: sinus 94, PR 152, QRS 86, QTc 432, **T waves normal** (K down). C. diff negative (1240).
Response tracker column (Vitals & Assessment pair)	108/66 (80); 94/60, HR 114, dizzy: POSITIVE; 92, T normal; 118; Na 132 (+6/9 h); K 4.9; alert, x3; about 260 mL/h; none; better, but Na too fast
What changed since the last stop	Better: BP 94/58 to 108/66, HR 112 to 92, alert x3, nausea gone, glucose stable, K 5.5 to 4.9, T waves normal. New concerns: positive orthostatics, sodium +6 in 9 h with about 260 mL/h of pale urine.
Actions to suggest (Vitals & Assessment + Meds & Treatments)	Fluid change when ordered (1530); neuro checks; urine every void; up with assist only; hydrocortisone 1600; next BMP 1700.
TEAM (charge): top 3 priorities	1) Call the provider now (SBAR): sodium +6 in 9 h with a water diuresis; ask for a fluid change and call limits. 2) Fall safety: up with assist only, dangle first, commode close, bed alarm (positive orthostatics, constant urge to void). 3) Hydrocortisone 50 mg IV at 1600 on time; keep strict I&O and neuro checks.

TEAM (charge): SBAR	**S:** Lauren Mitchell, 4W-22: her sodium went from 126 at 0406 to 132 at 1300, up 6 in 9 hours. She made 1,050 mL of pale urine in 4 hours, about 260 an hour. **B:** Adrenal crisis on hydrocortisone 50 mg every 6 hours and D5NS at 150. The order says no more than 8 in 24 hours. **A:** Alert x3, no neuro changes, but at this rate she will pass the limit tonight. Orthostatics positive: 108/66 to 94/60, HR 92 to 114, dizzy. **R:** Would you change the IV fluid or rate? When should I recheck the sodium, and at what sodium or urine output do you want a call? I am keeping her up with assist only.
Prediction (no wrong answers)	Sodium should level off after the fluid change; orthostatics should improve with fluids and steroid. What happens: Na 133 at 1700 (+1 in 4 h), orthostatics negative at 1900.
Charge nurse notes	Look for: each pair was asked its question, the charge nurse decided, a time check. Credit any honest notes.

Instructor as the provider (suggested reply): Thank you for catching that. Stop the D5NS and start D5 0.45% NaCl at 100 mL/h. Keep the sodium 134 or lower until 0400 tomorrow; call me for a sodium over 134, urine over 250 mL/h for 2 hours, or any new confusion, headache or trouble speaking or swallowing. Up with one assist only; repeat orthostatics at 1900. Give the 1600 hydrocortisone on time.

1900 Monday: Checkpoint 4 of 6

Monitor (the NOW row): HR 86, BP 112/70 lying (MAP 84), RR 16, SpO2 99% room air, temp 98.6, pain 1/10; rhythm: Normal sinus rhythm. **Standing (aloud):** 104/66, HR 96, no dizziness (negative). **Bedside glucose (aloud):** 142

Urine and I&O for the Vitals pair: 1500-1900: in 905 (IV 425, PO 480) / out 1,100: voids 1600 300, 1730 320, 1850 280 (900 mL) + loose stool 200 at 1640.

Portray: Alert, oriented x4, joking: "When can I go home? Ethan has a game on Thursday." Standing **104/66, HR 96**, no dizziness.

Situation (blanks with answers in brackets): It is **1900**, shift change: **your team now becomes the night shift**. At 1530 the provider changed the IV fluid from **[D5NS 150 mL/h]** to **[D5 0.45% NaCl 100 mL/h]**. Lauren has had ginger ale and juice, a **[regular]** diet was ordered at 1830, and she had one loose stool at 1640. The 1700 sodium is **[133]** and potassium **[4.4]**. The stool GI panel found **[norovirus]**, so she stays on **[contact (enteric)]** precautions; clean hands with **[soap and water]**. Orthostatic vitals were ordered again for 1900. Tonight's hydrocortisone is due at **[2200]** and **[0400]**.

Questions a strong charge nurse asks (not printed for students; prompt only if stuck): Meds & Treatments: "What fluid replaced D5NS at 1530 and why? Which doses are due tonight (times)? Is fludrocortisone still held, and why?" **Vitals & Assessment:** "Repeat orthostatics: positive or negative? Mental status, eating, nausea, urine. Is she safe to walk with help?" **Results & Labs:** "Sodium at 1700 against 1300 and 0406: did the fluid change work? Potassium trend? What did the GI panel find, and what does it mean for precautions?"

New results (chart values)

Test	Normal	Mon 0900	Mon 1300	Mon 1700
Sodium	136-145	128	132	133
Potassium	3.5-5.1	5.5	4.9	4.4
Creatinine	0.5-1.1	1.4	1.2	1.0
Glucose (serum)	70-110	74	96	112

Box	Model answer and rationale
MEDS & TREATMENTS pair	D5NS stopped at 1530; **D5 0.45% NaCl 100 mL/h** (half the sodium, lower rate) to slow the sodium rise. Hydrocortisone 50 mg IV due **2200 and 0400**: on time. Fludrocortisone still held (hydrocortisone 200 mg/day). Enoxaparin 40 mg SubQ due 2100. Regular diet with normal salt to taste, but **no extra salt or salty broth until 0400** (sodium-limit window); fluids to thirst.
VITALS & ASSESSMENT pair	HR 86, BP 112/70 lying (MAP 84), RR 16, SpO2 99%, T 98.6, pain 1/10, glucose 142. **Orthostatic: standing 104/66, HR 96, no dizziness = negative** (-8 / +10). Alert, oriented x4, joking, hungry, no nausea. Urine 1600 300, 1730 320, 1850 280 = 900 mL (about 225 mL/h) + loose stool 200 at 1640. In 905 / out 1,100.
RESULTS & LABS pair: new results and significance	**Sodium 133 at 1700**: +1 in 4 h after the fluid change (the plan is working); +7 since 0406 (13 h), still within the limit. Potassium 4.4 (falling), creatinine 1.0 (near baseline 0.8), glucose 112. **GI panel: norovirus detected** (C. diff negative): keep contact (enteric) precautions until 48 h after the diarrhea stops; soap and water (gel does not kill norovirus).
Response tracker column (Vitals & Assessment pair)	112/70 (84); 104/66, HR 96: negative; 86; 142; Na 133; K 4.4; x4; about 225 mL/h; none, eating; better
What changed since the last stop	Better on every measure: BP 108/66 to 112/70, orthostatics positive to negative, HR 92 to 86, x3 to x4, eating, Na rise slowed (132 to 133), K 4.9 to 4.4, creatinine 1.2 to 1.0. **Response: effective.**
Actions to suggest (Vitals & Assessment + Meds & Treatments)	Night plan: hydrocortisone 2200/0400; enoxaparin 2100; BMP 2100 and 0100; walk with assist; precautions; learning-needs check.
TEAM (charge): top 3 priorities	1) Evaluate and hand off with data: the treatment is working; keep the sodium limit (134 until 0400) and the hydrocortisone on time (2200, 0400). 2) Safe walking: assist, dangle first, call light; norovirus precautions with soap and water. 3) Start teaching now that she is alert: find out what she and Daniel know about sick days.
TEAM (charge): SBAR	**S:** Handoff, Lauren Mitchell, 4W-22, adrenal crisis day 1, now responding to treatment. **B:** Na 126 at 0406, 132 at 1300 (+6 in 9 h); fluid changed to D5 0.45% NaCl 100 at 1530; limit 134 until 0400. Low glucose at 1100 treated with D50. Norovirus detected. **A:** BP 112/70, orthostatics now negative, HR 86, alert x4, eating, Na 133, K 4.4, creatinine 1.0: treatment is working. **R:** Hydrocortisone 2200 and 0400 on time, BMP 2100, walk with assist, contact precautions, no extra salt tonight, start sick-day teaching.
Prediction (no wrong answers)	She will feel well but may have trouble sleeping on high-dose steroid; potassium keeps falling. What happens: "wired" at 2300, glucose 152, K 4.0.
Charge nurse notes	Look for: each pair was asked its question, the charge nurse decided, a time check. Credit any honest notes.

Instructor as the provider (suggested reply): (Night hospitalist) She is responding. Keep the fluid and the sodium limit as ordered, hydrocortisone at 2200 and 0400 on time. She can walk in the room and hall with assist. Contact precautions for norovirus. Regular diet, but no extra salt or broth until 0400.

2300 Monday night: Checkpoint 5 of 6

Monitor (the NOW row): HR 80, BP 116/72 lying (MAP 87), RR 16, SpO2 98% room air, temp 98.4, pain 0/10; rhythm: Normal sinus rhythm. **Standing (aloud):** 112/70, HR 88, no symptoms (negative). **Bedside glucose (aloud):** 152

Urine and I&O for the Vitals pair: 1900-2300: in 1,120 (IV 400, PO 720) / out 700: voids 2000 250, 2130 250, 2250 200.

Portray: "I'm wired. My mind won't stop. Who's going to take Ethan to school? Can I have something to knock me out?" Standing **112/70, HR 88**. Glucose **152**.

Situation (blanks with answers in brackets): It is **2300**. Lauren ate **[75]**% of her dinner at 2000. The 2100 sodium is **[133]** and potassium **[4.0]**; the sodium limit until 0400 is **[134]**. The hydrocortisone was due at **[2200]**: given? **[yes, on time]**. Her call light is on. "My mind won't stop. Who's going to take Ethan to school? Can I have something to knock me out?" Her bedside glucose is **[152]** (ask the instructor).

Questions a strong charge nurse asks (not printed for students; prompt only if stuck): Meds & Treatments: "Was the 2200 dose given? What can we offer for sleep, and which sleep medicines should she NOT get? Bedside glucose now: does she need insulin?" **Vitals & Assessment:** "Orthostatics (every 8 hours), mood and mental status. What is in place to keep her from falling tonight?" **Results & Labs:** "Sodium at 2100: still within the limit? Potassium from 6.0 to now: where is it heading, and why? When is the next check?"

New results (chart values)

Test	Normal	Mon 1300	Mon 1700	Mon 2100
Sodium	136-145	132	133	133
Potassium	3.5-5.1	4.9	4.4	4.0
Creatinine	0.5-1.1	1.2	1.0	0.9
Glucose (serum)	70-110	96	112	128

Box	Model answer and rationale
MEDS & TREATMENTS pair	Hydrocortisone 50 mg IV at 2200: given on time (check the MAR). Sleep: non-drug measures first (lights down, quiet, cluster care, talk about Ethan, call Daniel); what the provider orders at 2315: **melatonin 3 mg PRN**. **No diphenhydramine or benzodiazepines** (falls at night with a diuresis, an IV and confusion risk). Glucose 152: a steroid effect after dinner in a person without diabetes: **no insulin**; recheck per order, call if over 180 twice.
VITALS & ASSESSMENT pair	HR 80, BP 116/72 (MAP 87), RR 16, SpO2 98%, T 98.4, pain 0, glucose 152. Orthostatic: standing 112/70, HR 88, no symptoms = negative. Awake, restless, talkative, anxious about her son and work ("wired"); oriented x4. Urine 2000 250, 2130 250, 2250 200 (700 mL, about 175 mL/h). In 1,120 / out 700.
RESULTS & LABS pair: new results and significance	Sodium 133 (stable; limit 134 until 0400). Potassium 4.0 (6.0, 5.5, 4.9, 4.4, 4.0): falling with fluids, dextrose and hydrocortisone acting like aldosterone, plus stool losses: **expect it to go low**; next BMP 0100. Creatinine 0.9, glucose 128.
Response tracker column (Vitals & Assessment pair)	116/72 (87); 112/70, HR 88: negative; 80; 152; Na 133; K 4.0; x4, "wired"; about 175 mL/h; none; better (insomnia)
What changed since the last stop	Vitals better again (116/72, HR 80, orthostatics negative). New: insomnia and anxiety (steroid effect + worry), glucose 152 (steroid + dinner). Na steady at 133; K 4.4 to 4.0.
Actions to suggest (Vitals & Assessment + Meds & Treatments)	Quiet room, melatonin per order, bed alarm, scheduled toileting, BMP 0100, hydrocortisone 0400.
TEAM (charge): top 3 priorities	1) Safe sleep without sedatives: non-drug measures, then melatonin if ordered; no diphenhydramine or benzodiazepines. 2) Night fall prevention: bed alarm, call light, commode/hat close, non-skid socks, "call, don't fall". 3) Watch the potassium trend (0100 BMP) and keep the 0400 hydrocortisone on time.
TEAM (charge): SBAR	**S:** Lauren Mitchell, 4W-22: she cannot sleep and is asking for "something to knock me out". Glucose 152 after dinner. **B:** Adrenal crisis, now stable on hydrocortisone 50 mg IV every 6 hours (2200 given). Orthostatics negative, Na 133 (limit 134 until 0400), K falling, 4.0. **A:** Steroid insomnia and worry; glucose 152 is a steroid effect; a sedative would raise her fall risk at night. **R:** May I give melatonin instead of diphenhydramine or a benzodiazepine? What glucose do you want called? Please confirm no insulin.
Prediction (no wrong answers)	Any reasoned guess. What happens: at 0240 she walks to the bathroom alone and feels lightheaded (no fall); the right forearm IV infiltrates; K 3.7 at 0100.
Charge nurse notes	Look for: each pair was asked its question, the charge nurse decided, a time check. Credit any honest notes.

Instructor as the provider (suggested reply): No diphenhydramine and no benzodiazepines. I will order melatonin 3 mg as needed. The 152 is the steroid: no insulin; call me if she is over 180 twice. Keep the bed alarm on.

0300 Tuesday: Checkpoint 6 of 6

Monitor (the NOW row): HR 76, BP 118/74 lying (MAP 89), RR 14, SpO2 98% room air, temp 98.2, pain 0/10; rhythm: Normal sinus rhythm (T waves a little smaller). **Standing (aloud):** not repeated (just back to bed after the near-fall). **Bedside glucose (aloud):** 128

Urine and I&O for the Vitals pair: 2300-0300: in 640 (IV 400, PO 240) / out 600: voids 0030 250, 0245 200 (450 mL) + soft loose stool 150 at 0240.

Portray: Embarrassed: "I didn't want to bother anyone. I just got dizzy for a second." No injury. Right forearm IV: puffy, pale, cool, about 3 cm of swelling, feels "tight"; the pump has not alarmed. Glucose **128**.

Situation (blanks with answers in brackets): It is **0300 Tuesday**. At **[0240]** the bed alarm went off: Lauren had walked to the bathroom **[alone]**, had a soft loose stool, felt lightheaded and sat down on the toilet, then pulled the cord. You found her sitting there. **She did not fall.** She is back in bed. The 0100 potassium is **[3.7]** (it was 6.0 at 0406) and sodium **[133]**. Look at her and at both IV sites: the right forearm IV is **[puffy, pale, cool, swollen]**. The next hydrocortisone is due at **[0400]**.

Questions a strong charge nurse asks (not printed for students; prompt only if stuck): Meds & Treatments: "The right forearm IV: what do you see and what do you do? How will the 0400 hydrocortisone go in? Anything needed for the potassium?" **Vitals & Assessment:** "After the near-fall: neuro check, BP and HR lying then sitting, glucose, skin check. Is this the crisis coming back? Why or why not?" **Results & Labs:** "Potassium 6.0 then down to the 0100 value: why is it falling, and what is the next step? Magnesium and phosphorus? Sodium against the limit?"

New results (chart values)

Test	Normal	Mon 1700	Mon 2100	Tue 0100
Sodium	136-145	133	133	133
Potassium	3.5-5.1	4.4	4.0	3.7
Creatinine	0.5-1.1	1.0	0.9	0.9
Glucose (serum)	70-110	112	128	136
Magnesium	1.7-2.2			1.9
Phosphorus	2.5-4.5			2.9

Box	Model answer and rationale
MEDS & TREATMENTS pair	Right forearm 20 g IV: puffy, pale, cool, 3 cm swelling, feels tight = **infiltration**: stop the infusion, remove the IV, elevate the arm, compress per policy, mark and measure, chart the removal. Move the D5 0.45% NaCl to the **18 g left forearm** (keep a working IV: **hydrocortisone 50 mg IV due 0400** goes through it). Potassium 3.7: no replacement yet; recheck 0500 and replace if 3.5 or lower (0310 note).
VITALS & ASSESSMENT pair	HR 76, BP 118/74 (MAP 89), RR 14, SpO2 98%, T 98.2, pain 0, glucose 128, sinus rhythm (T waves a little smaller). Embarrassed ("I didn't want to bother anyone"); no head strike, no injury, oriented x4. Not an adrenal relapse: BP normal, alert, glucose normal, steroid on schedule; likely straining, getting up fast and night diuresis. Urine 0030 250, 0245 200 (450 mL) + soft stool 150 at 0240. In 640 / out 600.
RESULTS & LABS pair: new results and significance	Potassium **3.7** (6.0 to 3.7 in 21 h): fluids, dextrose and high-dose hydrocortisone (aldosterone-like) plus stool losses; still normal, but heading low, and fludrocortisone restarts soon. Magnesium 1.9, phosphorus 2.9 (normal). Sodium 133 (limit 134 until 0400). Creatinine 0.9, glucose 136.
Response tracker column (Vitals & Assessment pair)	118/74 (89); not repeated; 76; 128; Na 133; K 3.7; x4; about 113 mL/h; none; same / better (near-fall, K falling)
What changed since the last stop	Vitals the best so far (118/74, HR 76). New: near-fall in the bathroom, infiltrated right forearm IV, K 4.0 to 3.7.
Actions to suggest (Vitals & Assessment + Meds & Treatments)	IV moved; hydrocortisone 0400; commode, bed alarm, light on; event report; BMP and CBC 0500.
TEAM (charge): top 3 priorities	1) After the near-fall: full check, back to bed safely, tell the provider, safety (event) report, re-teach "call, don't fall", commode at the bedside. 2) Remove the infiltrated IV, keep one working IV, give the 0400 hydrocortisone on time. 3) Potassium falling: watch the rhythm, recheck at 0500, replace per order.
TEAM (charge): SBAR	**S:** Lauren Mitchell, 4W-22: at 0240 she walked to the bathroom alone, felt lightheaded and sat down on the toilet. No fall, no injury. **B:** Adrenal crisis day 1, improving; orthostatics negative at 1900 and 2300. On D5 0.45% NaCl at 100 through the right forearm IV. **A:** BP 118/74, HR 76, alert, glucose 128. The right forearm IV is infiltrated (3 cm cool swelling); I stopped the fluid. K 3.7 at 0100, down from 6.0. **R:** I am removing that IV and moving the fluid to the left forearm 18 g so the 0400 hydrocortisone can go in. Do you want anything for the potassium before the 0500 labs? Bed alarm and commode in place; I will file a safety report.

Prediction (no wrong answers)	Sodium near the limit and potassium near 3.5 by morning; she will be ready to switch to pills. What happens: Na 134, K 3.5 at 0500, KCl ordered, oral taper and fludrocortisone restart.
Charge nurse notes	Look for: each pair was asked its question, the charge nurse decided, a time check. Credit any honest notes.

Instructor as the provider (suggested reply): Take out the right IV and run the fluid through the left forearm 18 g; give the 0400 hydrocortisone through it. We will recheck the potassium at 0500 and replace it if it is 3.5 or lower. Commode at the bedside, bed alarm on, and please do the safety report.

0700 Tuesday: Final check (page 1 of 2)

Monitor (the NOW row): HR 74, BP 120/76 lying (MAP 91), RR 14, SpO2 99% room air, temp 98.2, pain 0/10; rhythm: Normal sinus rhythm. **Standing (aloud):** 116/74, HR 80 (negative). **Bedside glucose (aloud):** 112

Urine and I&O for the Vitals pair: 0300-0700: in 760 (IV 400, PO 360) / out 500: voids 0430 250, 0640 250. Weight 0600: 59.8 kg (58.2 on arrival).

Portray: "I feel like myself. I never want to go through that again. What do I do next time Ethan gets sick?" Daniel: "Show me the shot." Standing **116/74, HR 80**. Weight **59.8 kg**. Glucose **112**.

Situation (blanks with answers in brackets): It is **Tuesday 0700**. Lauren slept from about 0330. The 0500 sodium is **[134]** (it was 126 on arrival) and potassium **[3.5]**. Endocrinology has written the plan to switch to **[oral hydrocortisone]** pills, and **[fludrocortisone]** restarts at **[0800]**. She wants to go home today: "What do I do next time Ethan gets sick?" Daniel is back at the bedside: "Show me the shot. I'll be the one giving it if she can't." Her weight this morning is **[59.8]** kg (arrival 58.2 kg).

Questions a strong charge nurse asks (not printed for students; prompt only if stuck): Meds & Treatments: "The new steroid plan: which doses today, at what times? Why does fludrocortisone restart at 0800 today? What else is due this morning?" **Vitals & Assessment:** "Lying and standing vitals, weight against arrival, mental status, eating, skin color. Is she ready to go home?" **Results & Labs:** "Sodium change from 0406 Monday to 0500 Tuesday: within the limit? Potassium now and next? Hemoglobin and eosinophils: what changed and why?"

New results (chart values)

Test	Normal	Tue 0100	Tue 0500
Sodium	136-145	133	134
Potassium	3.5-5.1	3.7	3.5
Creatinine	0.5-1.1	0.9	0.8
Glucose (serum)	70-110	136	118
Magnesium	1.7-2.2	1.9	1.8

Test	Normal	Mon 0406	Tue 0500
Hemoglobin	12.0-15.5	14.6	11.9
WBC	4.0-11.0	10.8	12.6
Eosinophils	0-5	7.0	0.4

Box	Model answer and rationale
MEDS & TREATMENTS pair	**Oral taper** (order 0635): hydrocortisone 20 mg at 1000, 20 mg at 1500, 10 mg at 2000 today (50 mg/day; last IV dose was 0400), then 30 mg + 10 mg (double the home dose) until the diarrhea has stopped 24 h, then the home 15 mg + 5 mg. **Fludrocortisone 0.1 mg restarts 0800**: at 50 mg/day hydrocortisone no longer covers aldosterone. **KCl 20 mEq PO** due 0630 (whole, with food and water); BMP 1300. Levothyroxine 88 mcg due 0600 (resumed: "steroids first"). Saline lock at 1000 if drinking well; keep one IV until discharge.
VITALS & ASSESSMENT pair	HR 74, BP 120/76 lying (MAP 91), standing 116/74 HR 80 = negative; RR 14, SpO2 99%, T 98.2, pain 0, glucose 112. Alert, oriented x4, eating toast, no vomiting since 0440 Monday, last stool 0240 (soft). Weight 59.8 kg (58.2 on arrival: rehydrated). Urine 0430 250, 0640 250. Skin still tan (fades over months).
RESULTS & LABS pair: new results and significance	**Sodium 134** at 0500: 126 to 134 = **+8 in 25 h: at the limit, not over**; no neuro signs. **Potassium 3.5** (lower limit of normal; falling; fludrocortisone will lower it more): KCl 20 mEq and BMP 1300. Creatinine 0.8 (back to baseline). Hemoglobin 14.6 to 11.9: dilution after rehydration (albumin 4.6 to 3.6), not bleeding. WBC 12.6 with no fever and eosinophils 7.0% to 0.4%: steroid effect (cortisol is back). Blood cultures no growth at 24 h.
Response tracker column (Vitals & Assessment pair)	120/76 (91); 116/74, HR 80: negative; 74; 112; Na 134 (+8 in 25 h); K 3.5; x4; about 125 mL/h; none, eating; better: all criteria met
What changed since the last stop	Since 0300: steady and well. Since arrival: BP 98/60 to 120/76, orthostatics negative, HR 104 to 74, T waves normal, oriented x2 to x4, glucose stable without D50 since 1105 Monday, Na 126 to 134, K 6.0 to 3.5, creatinine 1.6 to 0.8, eating.
Actions to suggest (Vitals & Assessment + Meds & Treatments)	KCl, fludrocortisone, oral taper; teaching session with Daniel; case management; BMP 1300.
TEAM (charge): top 3 priorities	1) Steroid safety: oral hydrocortisone on time from 1000, fludrocortisone 0800, KCl now, BMP 1300. 2) Teach Lauren AND Daniel with teach-back: sick-day rules, the IM injection with return demonstration, medical ID and steroid card. 3) Discharge readiness: keeps the 1000 and 1500 doses down, eats, orthostatics negative, K normal at 1300; contact precautions continue.

TEAM (charge): SBAR	**S:** Lauren Mitchell, 46, day 2 of adrenal crisis; switching to oral steroid today, possible discharge this afternoon. **B:** Addison disease; crisis from norovirus with vomited steroid doses; no emergency kit at home. Last IV hydrocortisone 0400. **A:** Vitals normal, orthostatics negative, alert x4, eating. Na 134 (limit reached, not passed). K 3.5. Near-fall at 0240 without injury; right forearm IV removed; left forearm 18 g with D5 0.45% at 100. **R:** KCl now, fludrocortisone 0800, oral hydrocortisone 20 mg at 1000, saline lock at 1000 if drinking, BMP 1300, teaching with return demonstration for both before discharge. Contact precautions continue.
Charge nurse notes	Look for: each pair was asked its question, the charge nurse decided, a time check. Credit any honest notes.

Instructor as the provider (suggested reply): (Day team / endocrinology plan) Give the KCl now, fludrocortisone at 0800, oral hydrocortisone 20 mg at 1000, saline lock at 1000 if she is drinking, BMP at 1300. Teach both of them with a return demonstration before discharge; case management is arranging the kits.

Evaluation table (final page 1): model

Criterion	Goal	At arrival (Mon 0700)	Now (Tue 0700)	Met?
Blood pressure / MAP	SBP 100 or more, MAP 65 or more	98/60 (73)	120/76 (91)	Yes
Orthostatic vitals	Negative	not safe to stand	116/74, HR 80: negative	Yes
Heart rate and T waves	Under 100; T waves normal	104, tall peaked T	74, normal T (ECG 1330)	Yes
Glucose	70-180 without dextrose pushes	88 after D50 (49 in the ED)	112; no D50 since 1105 Mon	Yes
Sodium	Up, but no more than 8 in 24 h	126	134 (+8 in 25 h)	Yes, at the limit
Potassium	3.5-5.1; normal T waves	6.0	3.5 (falling: KCl)	Yes, but replace
Mental status	Oriented x4 (baseline)	drowsy, oriented x2	oriented x4	Yes
Kidneys and urine	Creatinine back to baseline (0.8); urine 0.5 mL/kg/h or more	creatinine 1.6; dark urine	creatinine 0.8; about 125 mL/h	Yes
Nausea / eating	No nausea, eating and drinking	nauseated, NPO	eating toast, no vomiting since 0440 Mon	Yes

Verdict (model): The treatment is effective: BP normal with negative orthostatics, HR 74 with normal T waves, oriented x4, glucose stable without dextrose since 1105 Monday, eating, creatinine back to 0.8, and the sodium corrected to 134 without passing the limit. One loose end: potassium 3.5 and falling (KCl now, BMP 1300).

Final page 2: teaching and discharge (model answers)

Box	Model answer
Handoff to the day team (SBAR), priorities, what changed, actions to suggest	Use the SBAR, priorities, "what changed" and actions on the previous page.
Meds pair: the steroid plan at home	**Oral taper** (order 0635): hydrocortisone 20 mg at 1000, 20 mg at 1500, 10 mg at 2000 today (50 mg/day; last IV dose was 0400), then 30 mg + 10 mg (double the home dose) until the diarrhea has stopped 24 h, then the home 15 mg + 5 mg. **Fludrocortisone 0.1 mg restarts 0800**: at 50 mg/day hydrocortisone no longer covers aldosterone.
Sick-day rules, the emergency injection, medical ID	(1) **Sick-day rules:** fever 100.4 F (38 C) or more, or sick enough for bed rest or antibiotics: **double** the daily hydrocortisone; fever 102.2 F (39 C) or more: **triple**; until 24-48 h after recovery; call the clinic. Vomit a dose within 30 minutes: repeat it. (2) **Inject 100 mg IM** (outer thigh, through clothing if needed) if she vomits again, cannot keep pills down, has severe diarrhea or is faint/confused, **then call 911**. Act-O-Vial: press, swirl, draw 2 mL, 90 degrees. **Return demonstration by Lauren and Daniel**; one kit at home, one in her bag; check expiry every 6 months. (3) **Medical alert ID** + steroid emergency card; tell every provider, dentist, the school nurse; surgery and procedures need extra steroid. Never skip or stop hydrocortisone; extra tablets everywhere.
Teach-back question	For example: "Tell me what you will do if Ethan's stomach bug comes back and you vomit your morning dose twice." (Model: take the dose again once if vomited within 30 minutes; if she vomits again: Daniel gives 100 mg IM in the outer thigh and calls 911.) Ask Daniel too: "Show me how you would give the shot."
Is she ready to go home?	Per endocrinology: keeps the 1000 and 1500 oral doses down, eats lunch, orthostatics negative, 1300 potassium normal, teach-back and return demonstration done by both, two kits dispensed, ID ordered, follow-up booked (endocrinology 1-2 weeks).
The story of the stay (for the debrief)	Arrival in crisis (BP 98/60, Na 126, K 6.0, glucose 49 in the ED); second low glucose 58 at 1100; better by 1500 but sodium +6 in 9 h and positive orthostatics, fluid changed; steady evening (orthostatics negative, eating); steroid insomnia; night near-fall and IV infiltration with K falling; ready for pills and teaching at 0700.

Response tracker: the whole model (page 3)

Measure	Mon 0700	Mon 1100	Mon 1500	Mon 1900	Mon 2300	Tue 0300	Tue 0700
BP lying (MAP)	98/60 (73)	94/58 (70)	108/66 (80)	112/70 (84)	116/72 (87)	118/74 (89)	120/76 (91)
BP and HR standing	standing not safe	not done	94/60, HR 114, dizzy: POSITIVE	104/66, HR 96: negative	112/70, HR 88: negative	not repeated	116/74, HR 80: negative
Heart rate / T waves	104, tall T	112	92, T normal	86	80	76	74
Glucose (bedside)	88	58 (142 after D50)	118	142	152	128	112
Sodium (latest)	Na 126	Na 128	Na 132 (+6/9 h)	Na 133	Na 133	Na 133	Na 134 (+8 in 25 h)
Potassium (latest)	K 6.0	K 5.5	K 4.9	K 4.4	K 4.0	K 3.7	K 3.5
Mental status	drowsy, x2	foggy, x2	alert, x3	x4	x4, "wired"	x4	x4
Urine output	150 mL in the ED, dark	about 95 mL/h	about 260 mL/h	about 225 mL/h	about 175 mL/h	about 113 mL/h	about 125 mL/h
Nausea / eating	nauseated	mild nausea	none	none, eating	none	none	none, eating
OVERALL	(start)	worse (glucose); labs better	better, but Na too fast	better	better (insomnia)	same / better (near-fall, K falling)	better: all criteria met

Scoring guide (20 points, three levels)

Give ONE group score from the shared packet (group score sheet, next page) plus an individual check (/4) for each of the 7 students. Be generous: complete and thoughtful earns full credit.

Category	Full credit	Good	Getting there
Teamwork and charge-nurse rotation (5)	5. A charge nurse is named at every stop, everyone was charge nurse once, and each pair reported its job when the charge nurse asked.	3 to 4. The rotation was followed most of the time. Most teammates took part.	1 to 2. The rotation was skipped, or one person did most of the work.
Clinical thinking at each checkpoint (6)	6. Every checkpoint has the pair boxes filled with real chart numbers and a reason (why it happens, and is the treatment working?).	4 to 5. Most checkpoints are filled in with numbers. One or two are thin.	1 to 3. Many boxes are blank or have one-word answers.
Safety, priorities and SBAR (5)	5. At 1100, 1500, 1900, 2300 and 0300: three priorities in order, a complete SBAR (all four parts) and a next step that keeps Lauren safe.	3 to 4. Most stops have priorities and an SBAR. A part is sometimes missing.	1 to 2. Priorities or SBAR are missing at many stops.
Completeness and reflection (4)	4. All boxes tried, the response tracker is filled at every stop, the final evaluation is done, and each person handed in a reflection.	2 to 3. Most boxes tried. The tracker, the evaluation or a reflection is missing.	1. Many blanks. Little reflection.

How the points add up

Category	Where the points come from
Teamwork and charge-nurse rotation (5)	Charge nurse named for all 7 stops = 2. Everyone had a turn as charge = 1. Every pair reported at the checkpoints and the peer checks were handed in = 2.
Clinical thinking at each checkpoint (6)	1 point for each checkpoint: 0700, 1100, 1500, 1900, 2300, 0300. Half a point for a mostly complete page.
Safety, priorities and SBAR (5)	1 point for each of 1100, 1500, 1900, 2300 and 0300 with 3 priorities and a 4-part SBAR. The 0700 SBAR is a warm-up (not scored).
Completeness and reflection (4)	Boxes tried (including the final 0700 pages) = 2. Response tracker filled at all 7 stops = 1. Individual reflections handed in = 1.

Examples

Category	Example of full credit	Example of partial credit
Clinical thinking (1500)	Results box: "Na 126 to 132 = +6 in 9 h; limit 8/24 h; too fast. Cortisol 1.8 + ACTH 1,240 = primary AI (gland), explains tan skin and fludrocortisone." Vitals box: "standing 94/60 HR 114, dizzy = positive."	"Sodium better. Cortisol low." (no numbers, no rate, no meaning): give half.
Safety and SBAR (1100)	Priorities: D50 now and recheck in 15 min; check the 1000 hydrocortisone; fall precautions. SBAR with all 4 parts and a clear request (glucose checks, ondansetron, come and see her).	Juice for a nauseated, foggy patient, or an SBAR with no recommendation: 0 for that stop.
Safety and SBAR (2300)	"No diphenhydramine or benzodiazepines; melatonin and non-drug sleep measures; 152 is a steroid effect, no insulin; bed alarm."	Asks for a sleeping pill without naming the fall risk.
Completeness	Tracker filled at all 7 stops, the evaluation table done with arrival and now numbers and a verdict.	Tracker missing 3 stops: lose the tracker point.

Reflection and peer check

- Reflection (3 boxes): 2 points on the individual sheet for honest answers with a reason (1 if thin). It also feeds the group "reflections handed in" point.
- Peer check: 1 point for handing it in. Use it to spot a team where one person did most of the work; read comments privately.
- If the team did not rotate the charge nurse, cap Teamwork at 3 and ask why in the debrief.

GROUP SCORE SHEET

Team name: _____ Date: _____ (one group score, then 7 individual checks)

Checkpoint check (tick what you see)

Page	Charge nurse named	Pair boxes with numbers	3 priorities	SBAR complete	Tracker column	Prediction
0700 (SBAR warm-up)	☐	☐	☐	☐	☐	☐
1100	☐	☐	☐	☐	☐	☐
1500	☐	☐	☐	☐	☐	☐
1900	☐	☐	☐	☐	☐	☐
2300	☐	☐	☐	☐	☐	☐
0300	☐	☐	☐	☐	☐	☐
Final 0700	☐	☐	☐	☐	☐	Evaluation ☐

Rubric

Category	Full	Good	Getting there	Points	Notes
Teamwork and charge-nurse rotation (5)	☐ 5	☐ 3-4	☐ 1-2	/ 5	
Clinical thinking at each checkpoint (6)	☐ 6	☐ 4-5	☐ 1-3	/ 6	
Safety, priorities and SBAR (5)	☐ 5	☐ 3-4	☐ 1-2	/ 5	
Completeness and reflection (4)	☐ 4	☐ 2-3	☐ 1	/ 4	
GROUP TOTAL				/ 20	

Individual check (7 students)

Student	Charge turn (1)	Reflection (2)	Peer check (1)	Individual /4
_____	☐	☐ 2 ☐ 1	☐	
_____	☐	☐ 2 ☐ 1	☐	
_____	☐	☐ 2 ☐ 1	☐	
_____	☐	☐ 2 ☐ 1	☐	
_____	☐	☐ 2 ☐ 1	☐	
_____	☐	☐ 2 ☐ 1	☐	
_____	☐	☐ 2 ☐ 1	☐	

Comments for the team:

Debrief questions (about 10 minutes; pick 8 to 10)

From the instructor guide, section 8. Put the trends on the board first: Na 126, 128, 132, 133, 133, 133, 134; K 6.0, 5.5, 4.9, 4.4, 4.0, 3.7, 3.5; SBP 98, 94, 108, 112, 116, 118, 120; glucose 88, 58 (142), 118, 142, 152, 128, 112.

#	Topic	Question
1	Charge nurse	What did the charge nurse do that helped the team most? What was hard about asking each pair and then deciding?
2	Cause	Why did a stomach bug make Lauren critically ill when her son only felt sick?
3	Hormones	Which findings at 0700 come from missing cortisol, which from missing aldosterone, and which from high ACTH? (Use your hormone map.)
4	Steroid	Why is fludrocortisone held on Monday and restarted Tuesday at 0800? What would happen if the 1600 hydrocortisone were held because "her BP is fine now"?
5	Glucose	Why did she become hypoglycemic at 1100 while on D5NS, and why D50 instead of juice?
6	Sodium	How did you know the sodium was rising too fast at 1500, and why does treatment of adrenal crisis cause it? What harm can a fast correction cause, and when?
7	Evaluation	Look at your response tracker. Which numbers prove the treatment is working, and at which stop could you first say so?
8	Potassium	Why did potassium fall from 6.0 to 3.5, and why does it matter more on Tuesday morning?
9	Safety	Lauren nearly fell at 0240 when she was "better". What made her high risk at night? What would prevent it?
10	Teaching	What exactly will you teach Lauren and Daniel, and how will you know they can do it at 2 a.m. when she is vomiting?

Ideas for good answers

- Charge nurse: asked each pair by name, kept time, made sure the tracker column was filled, kept the SBAR short and specific.
- Cause: illness raises the cortisol need 2-10 times; her gland cannot answer; vomiting removed her only source; diarrhea and vomiting lost salt and water on top of no aldosterone. No sick-day rules, no injection.
- Hormones: cortisol = low BP, glucose 49-58, nausea, belly pain, confusion, eosinophils 7%; aldosterone = Na 126, K 6.0 with peaked T, urine Na 48, dehydration, creatinine 1.6, salt craving; ACTH 1,240 = tan skin, palm creases, gums.
- Fludrocortisone: over about 50 mg/day hydrocortisone saturates the kidney enzyme and acts on the aldosterone receptor, so 200 mg/day covers both; at 50 mg/day oral it does not, so fludrocortisone restarts. Holding hydrocortisone brings the crisis back.
- Sodium: 126 to 132 = +6 in 9 h with about 260 mL/h pale urine (cortisol and volume back = ADH off = water diuresis). Osmotic demyelination 2-6 days later (speech, swallowing, weakness). Fix: lower-sodium fluid (D5 0.45% NaCl), D5W or desmopressin if needed.
- Evaluation: BP, orthostatics, HR and T waves, glucose, mental status, urine, Na and K: effective by 1900 (orthostatics negative, x4, eating, Na 133, K 4.4, creatinine 1.0).
- Potassium: fluids, dextrose (insulin moves K into cells), high-dose hydrocortisone acting like aldosterone, stool losses; fludrocortisone at 0800 lowers it further: KCl 20 mEq and BMP 1300.
- Teaching: double for 100.4 F or illness, triple for 102.2 F, inject 100 mg IM and call 911 for repeated vomiting or confusion, return demonstration by both, ID, card, extra tablets.

Take-aways to close with

- Known adrenal insufficiency + vomiting = emergency; steroid first, tests never delay it.
- Hydrocortisone is time-critical: on time, never held for "good" numbers.
- High-dose hydrocortisone covers aldosterone too; fludrocortisone comes back at 50 mg/day.
- Evaluate with data: BP and orthostatics, HR and T waves, glucose, mental status, urine, sodium and potassium.
- A rising sodium can be dangerous: count the change from the first value and the hours; watch the urine.
- Falls happen when patients feel better. Teach to prevent the next crisis.